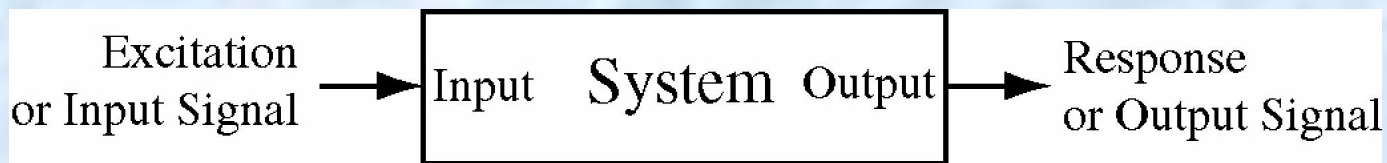
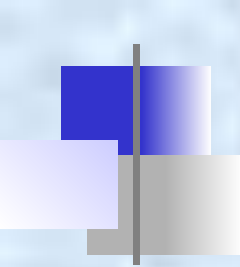


Introduction to Signals and Systems

Signals and Systems Defined

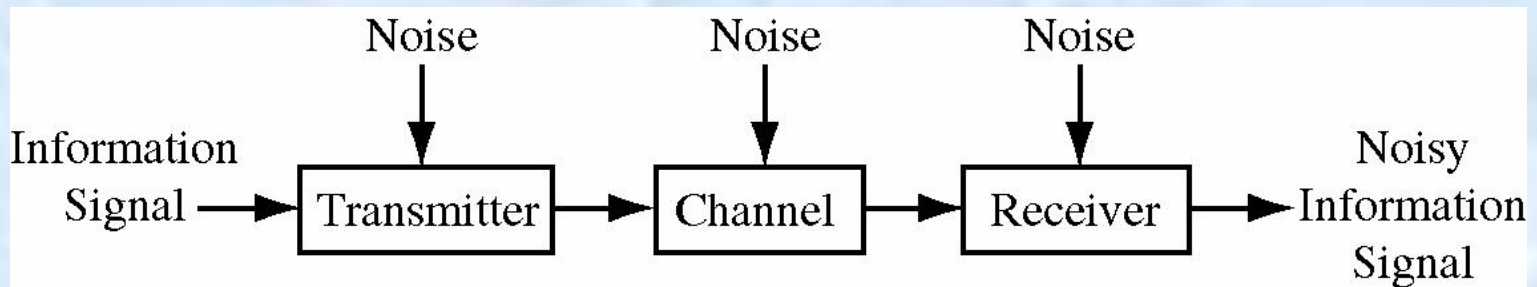
- A **signal** is any physical phenomenon which conveys information
- **Systems** respond to signals and produce new signals
- **Excitation** signals are applied at system **inputs** and **response** signals are produced at system **outputs**



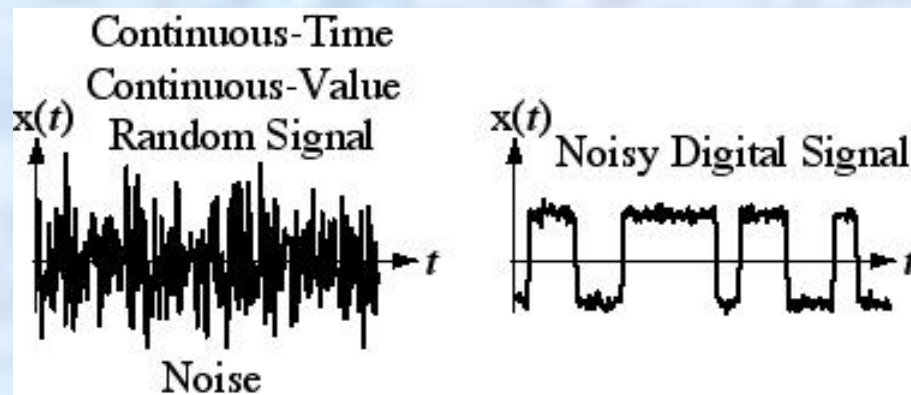
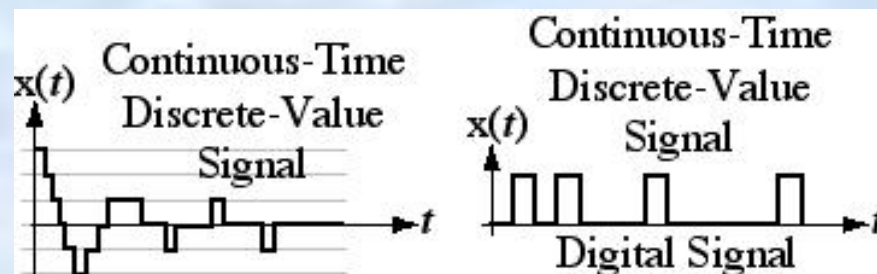
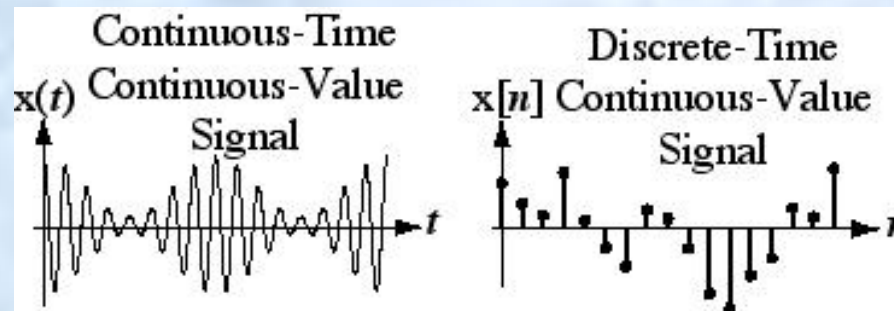


A Communication System as a System Example

- A communication system has an information signal plus **noise** signals
- This is an example of a system that consists of an interconnection of smaller systems



Signal Types

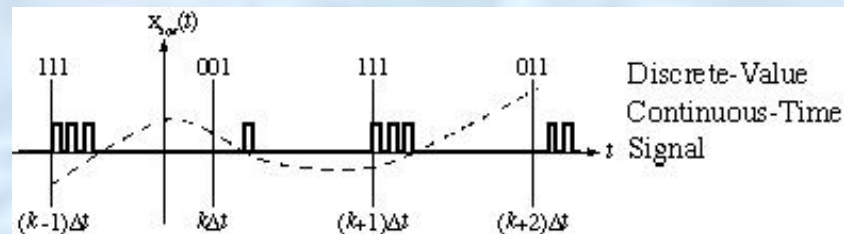
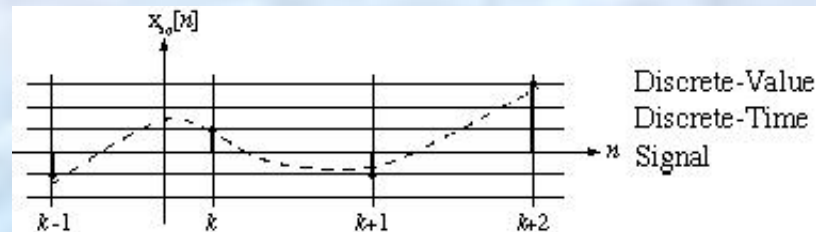
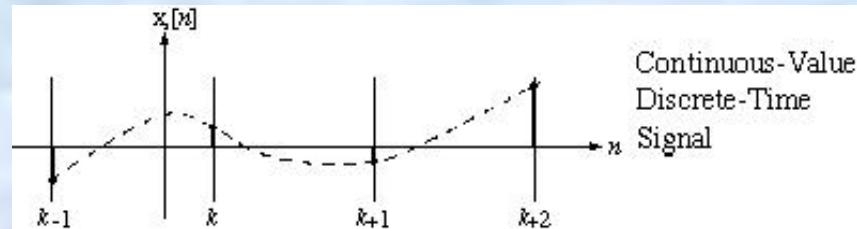
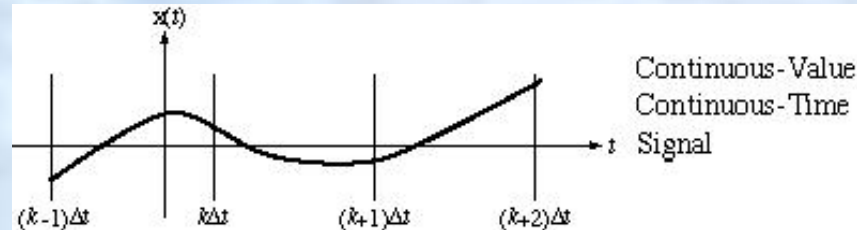


Conversions Between Signal Types

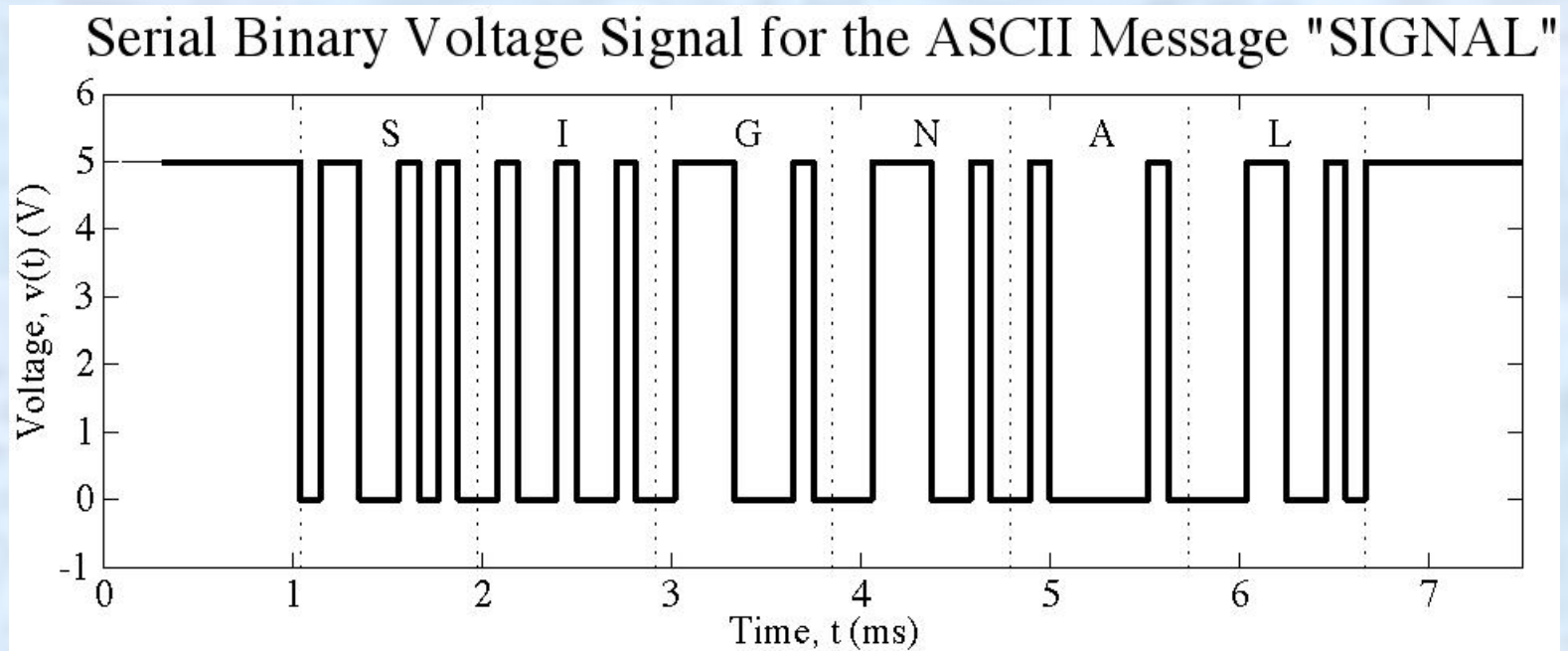
Sampling

Quantizing

Encoding

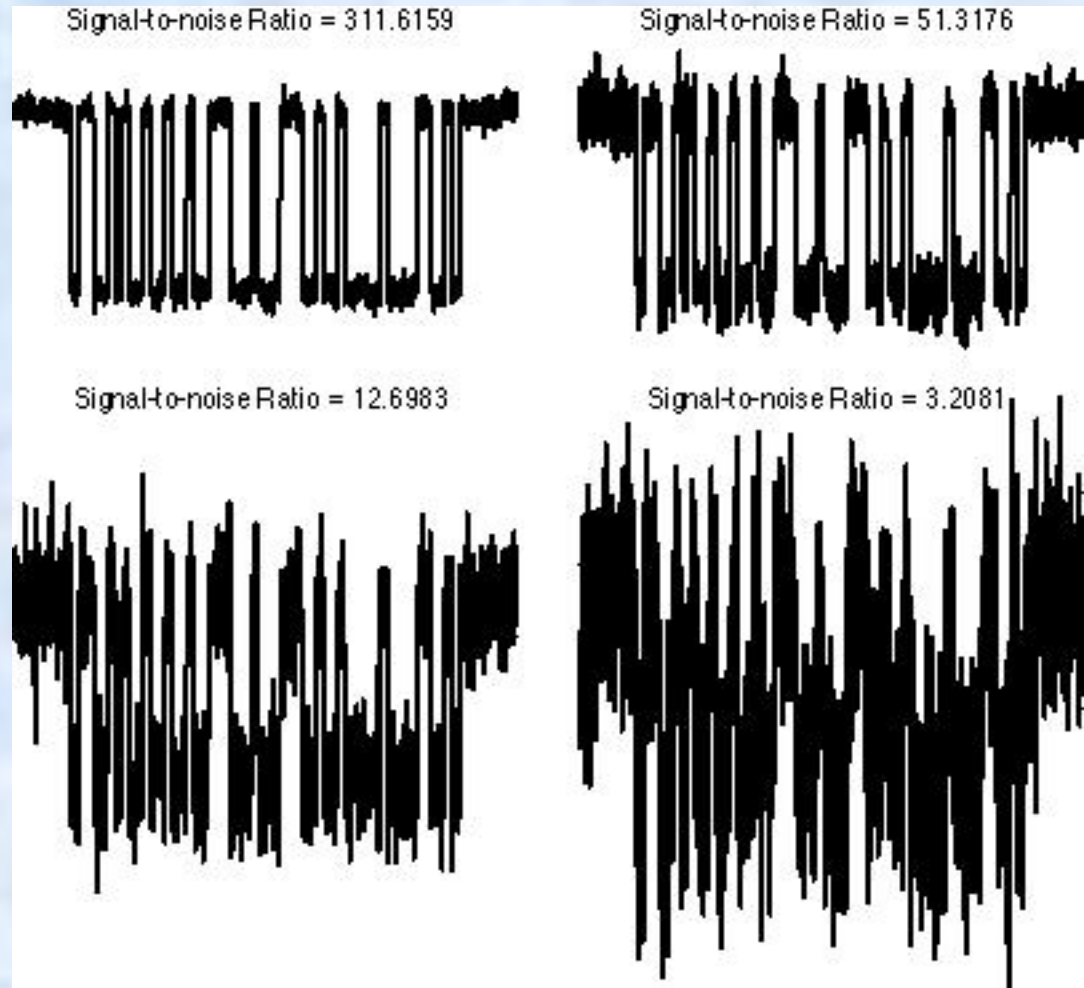


Message Encoded in ASCII



Noisy Message Encoded in ASCII

Progressively
noisier
signals



Bit Recovery in a Digital Signal Using Filtering

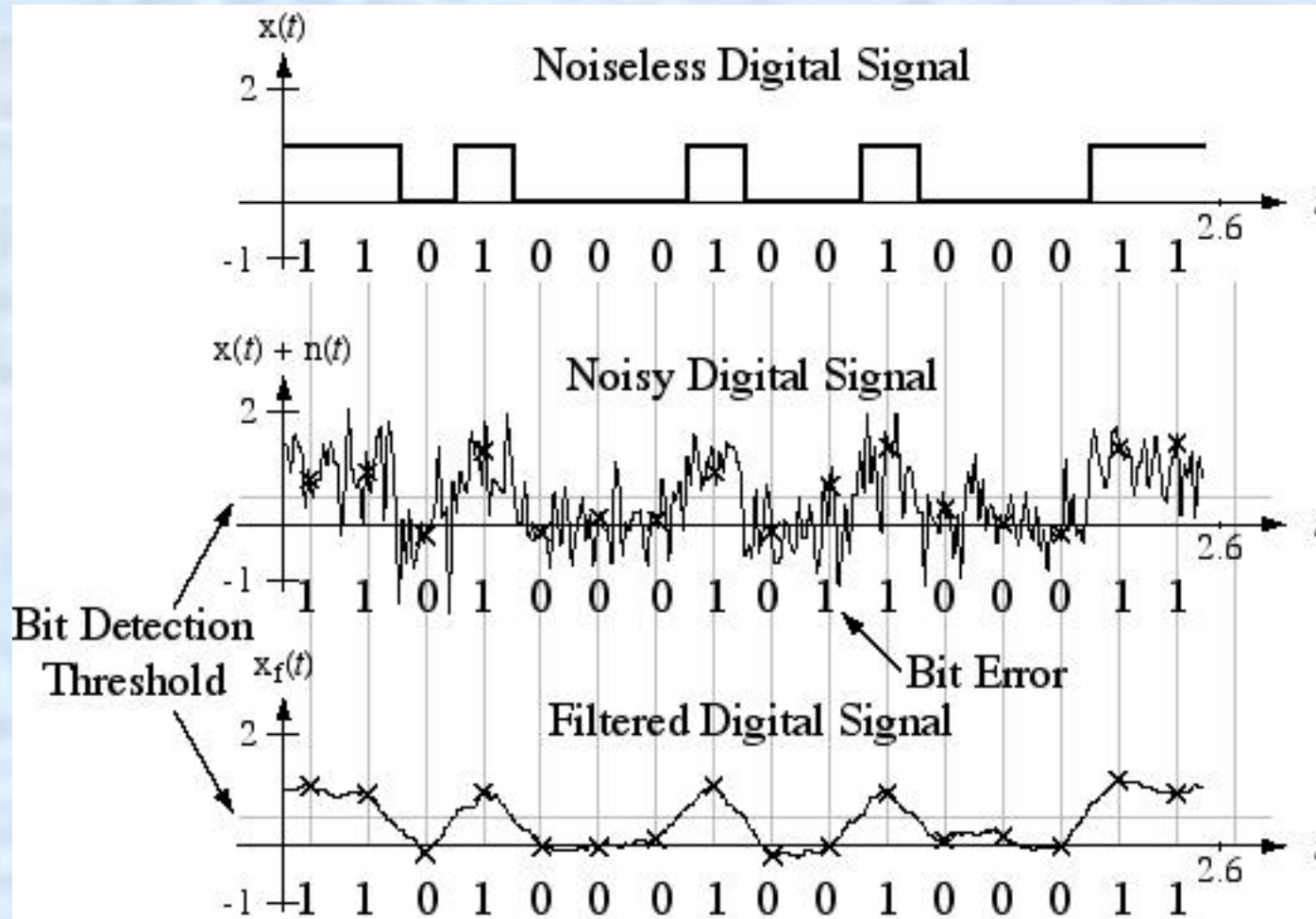




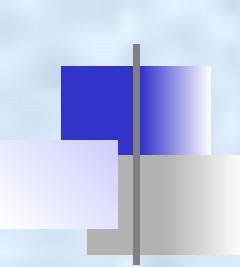
Image Filtering to Aid Perception

Original X-Ray Image



Filtered X-Ray Image



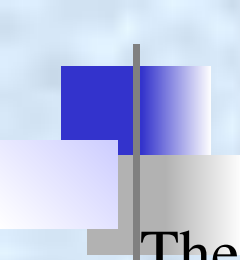


Discrete-Time Systems

In a **discrete-time** system events occur at points in time but not between those points. The most important example is a digital computer. Significant events occur at the end of each clock cycle and nothing of significance (to the computer user) happens between those points in time.

Discrete-time systems can be described by **difference** (not differential) equations. Let a discrete-time system generate an excitation signal $y[n]$ where n is the number of discrete-time intervals that have elapsed since some beginning time $n = 0$. Then, for example a simple discrete-time system might be described by

$$y[n] = 1.97y[n-1] - y[n-2]$$



Discrete-Time Systems

The equation

$$y[n] = 1.97 y[n-1] - y[n-2]$$

says in words

“The signal value at any time n is 1.97 times the signal value at the previous time $[n-1]$ minus the signal value at the time before that $[n-2]$.”

If we know the signal value at any two times, we can compute its value at all other (discrete) times. This is quite similar to a second-order differential equation for which knowledge of two independent initial conditions allows us to find the solution for all time and the solution methods are very similar.

Discrete-Time Systems

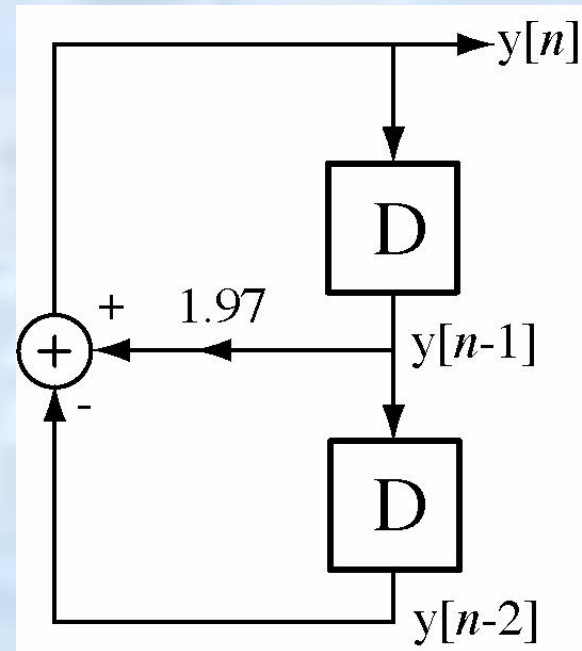
$$y[n] = 1.97y[n-1] - y[n-2]$$

We could solve this equation by iteration using a computer.

```
yn = 1 ; yn1 = 0 ;           ← Initial Conditions
while 1,
    yn2 = yn1 ; yn1 = yn ; yn = 1.97*yn1 - yn2 ;
end
```

We could also describe the system with a **block diagram**.

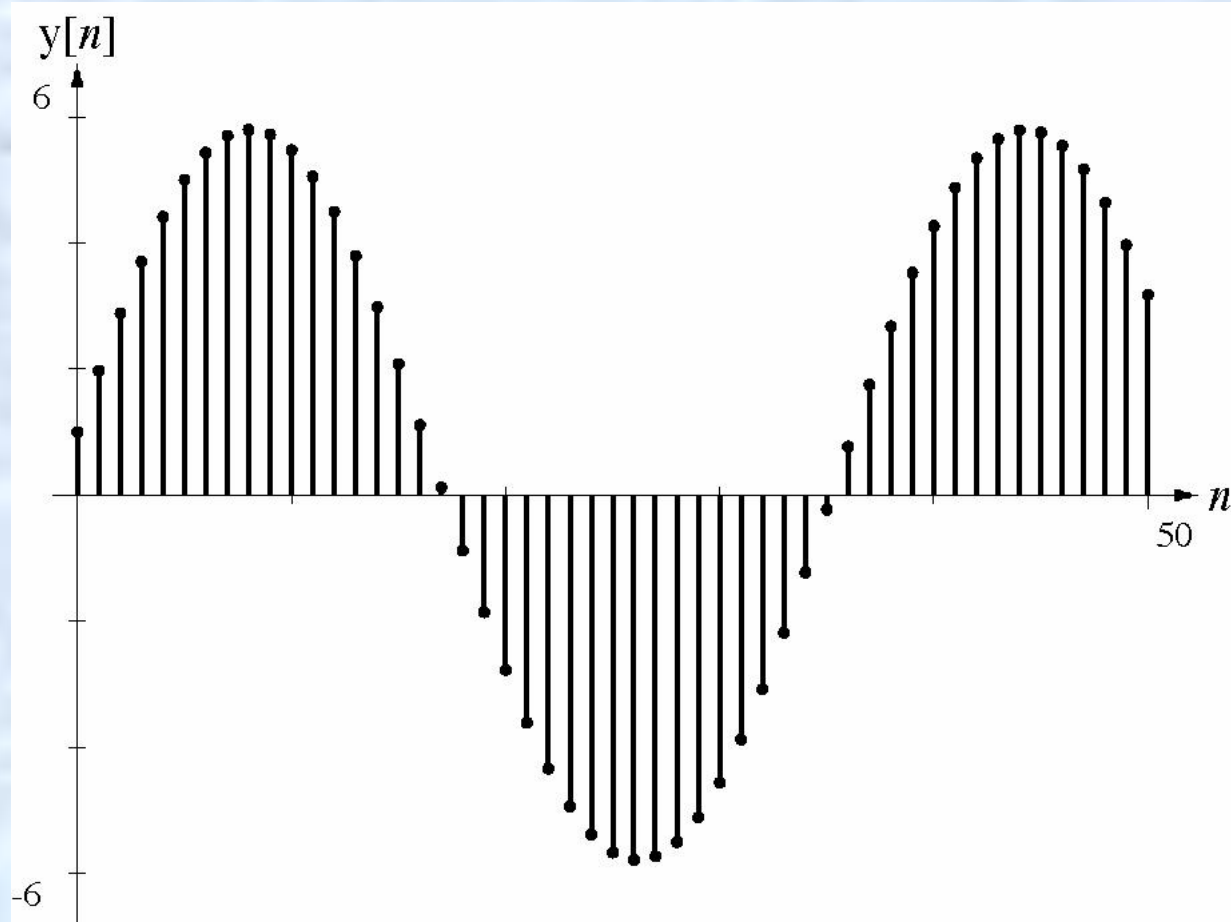
(“D” means delay one unit in discrete time.)

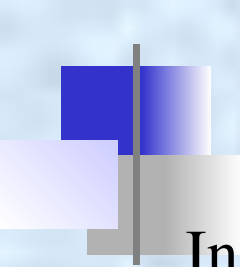


Discrete-Time Systems

$$y[n] = 1.97y[n-1] - y[n-2]$$

With the initial conditions $y[1] = 1$ and $y[0] = 0$ the response is





Feedback Systems

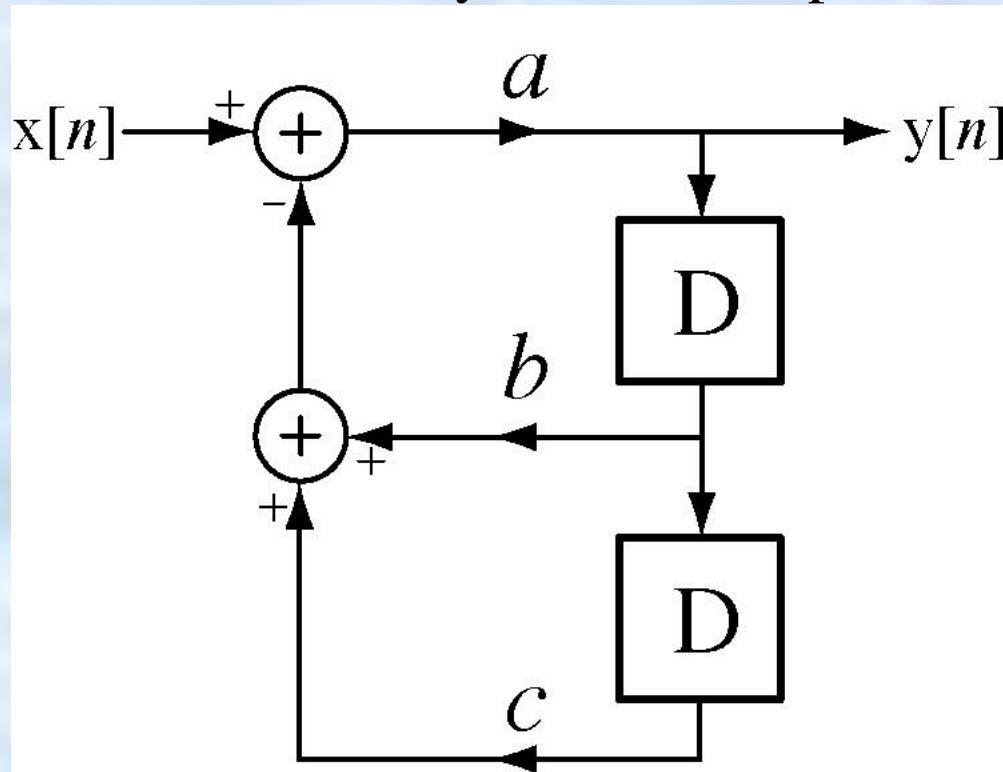
In a **feedback** system the response of the system is “fed back” and combined with the excitation in such a way as to optimize the response in some desired sense. Examples of feedback systems are

1. Temperature control in a house using a thermostat
2. Water level control in the tank of a flush toilet.
3. Pouring a glass of lemonade to the top of the glass without overflowing.
4. A refrigerator ice maker that keeps the bin full of ice but does not make extra ice.
5. Driving a car.

Feedback systems can be continuous-time or discrete-time or a mixture of the two.

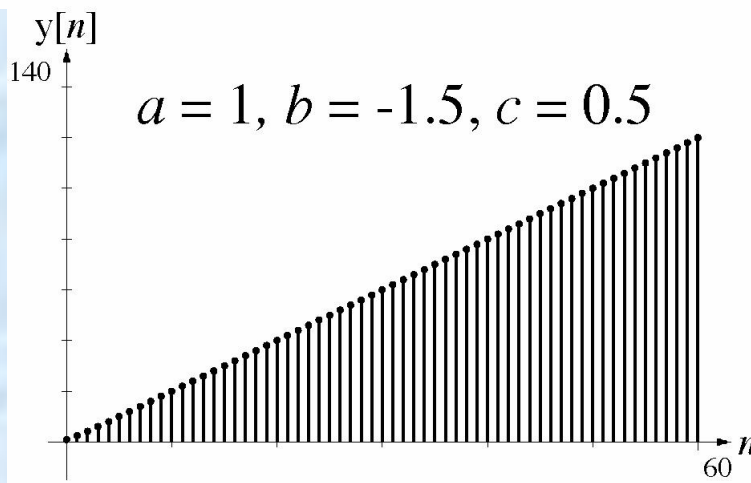
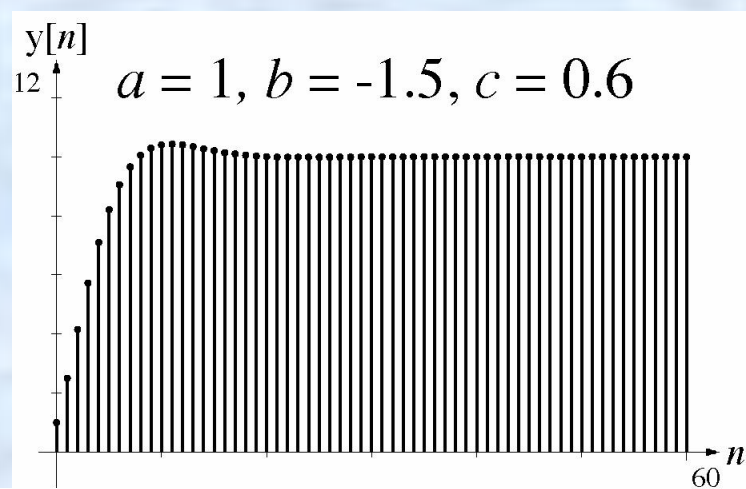
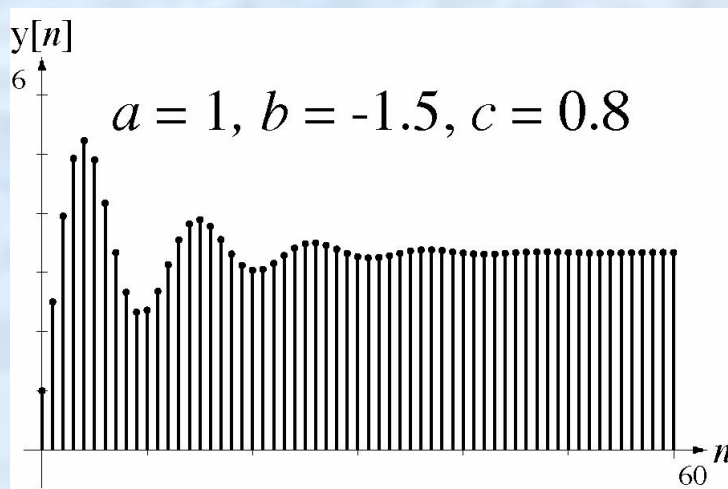
Feedback Systems

Below is an example of a discrete-time feedback system. The response $y[n]$ is fed back through two delays and gains b and c and combined with the excitation $x[n]$. Different values of a , b and c can create dramatically different responses to the same excitation.

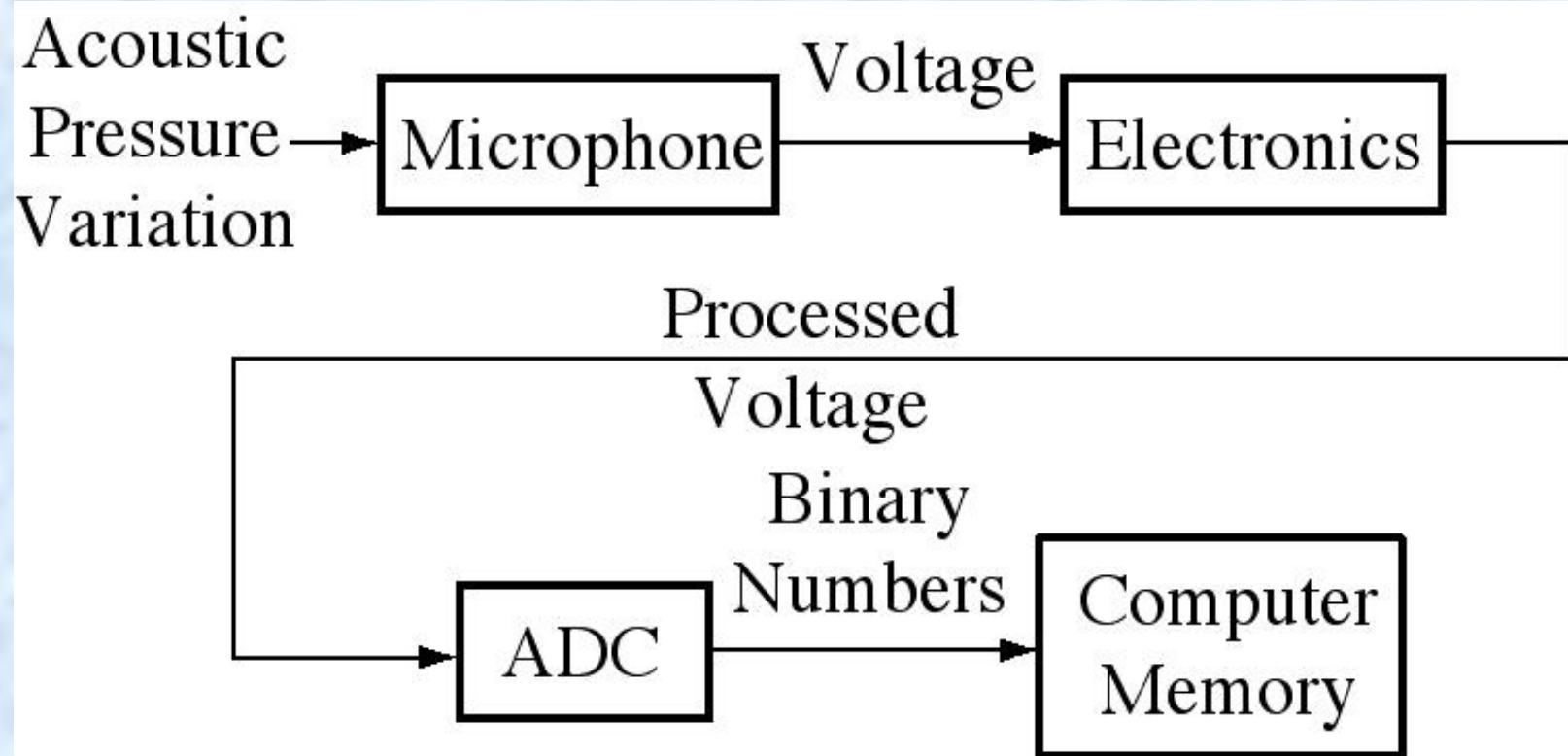


Feedback Systems

Responses to an excitation that changes from 0 to 1 at $n = 0$.



Sound Recording System



Recorded Sound as a Signal Example

- “s” “i” “gn” “al”

